UNIVERSITY OF LEICESTER

Properties of Materials
Lecture 1

Dr. Alkis Tsamis
at441@leicester.ac.uk

CTGU/UoL JEP, September 2024

UNIVERSITY OF LEICESTER

Course Topics and Schedule

Lecture 1: Module Overview, Module Aim, Why Study Materials, The World of Materials (1), History - The Evolution of Materials, Materials Selection in Design (1), Properties of Engineering Materials (1), Products, Three-Way Relationship (1), Origins of Material Properties, How Do We - As Engineers - Select Materials
Lecture 2: Concepts and Terminology (1), Properties of Engineering Materials (2), Mechanical Properties (1), Elastic Properties, Elastic Properties: Covalent and Metallic Bonds - Ionic Model - The Origin of Modulus, Plastic Properties, Fracture / Fatigue / Creep, Thermal Properties (1), Density and Corrosion, Corrosion: Anodic and Cathodic Reactions, Ellingham Diagram
Lecture 3: Structure-Property-Processing Relationship, Structure Sensitive Properties, Structure Insensitive Properties, The Three-Part Relationship, Materials Property Web Sites, Materials Selection, Material Property - Charts: Modulus - Density, Mechanical Properties (2), Thermal Properties (2), A TOOL for Materials Selection, Structure Sensitivity, Structure Sensitivity in Metals, Controlling Structure, Case Study

UNIVERSITY OF LEICESTER

Course Topics and Schedule

Lecture 4: Availability of Materials, Case Study, Exam Question 2013, Example Questions, The World of Materials (2), Materials Selection in Design (2), Properties of Engineering Materials (3), Three-Way Relationship (2), Learning Objectives, Why Do We Need a Selection Methodology, Mechanical Properties (3), Thermal Properties (3), Trade-Offs Must Be Made, Materials Selection in Design (3), Materials Selection Process (1)
Lecture 5: Step 1 - Translation, Step 2 - Screening, Materials Selection Process (2), Step 3 - Ranking Materials, Example - Hydraulic Cylinder Tie Rods, Step 4 - Refining Selection, Concepts and Terminology (2)
Lecture 6: The Elastic Moduli, Deformation of Brittle Materials, Deformation of Ductile Materials, Hooke's Law - Young's Modulus, Linear Elastic Behaviour, Structural Engineering, What Might Have Saved the Bridge, Engineering Stress and Strain, True Stress and Strain, Comparison
Lecture 7: Practice Time, Definitions, Concepts and Terminology (3), Task
Exercise Seminar 1

UNIVERSITY OF LEICESTER

Course Topics and Schedule

Lecture 8: Bonding Between Atoms, Learning Objectives, Bonding Energies, Primary Bonding, Ionic Bonding, Covalent Bonding, Metallic Bonding, Crystal Structures, Body-Centred Cubic, Face-Centred Cubic, Secondary Bonding, Van der Waals Forces, Hydrogen Bonding, Bonding Energies and Properties, Bonding Types and Properties, Concepts and Terminology (4)
Lecture 9: Recap from Last Time, Task
Exercise Seminar 2 / Lecture 10: Packing of Atoms in Solids (1), Learning Objectives, The Structure of Metals and Ceramics (1), Unit Cell Parameters, Face-Centred Cubic (fcc), Hexagonal Close-Packed (hcp), Crystal Structures - Metals, Crystal Structures - Ceramics, Concepts and Terminology (5)
Lecture 11: Packing of Atoms in Solids (2), Learning Objectives, Atomic Packing, Atomic Packing - bcc Structures, Atomic Packing Factor - bcc, Atomic Packing - fcc Structures, Atomic Packing Factor - fcc, Atomic Packing - hcp Structures, Atomic Packing Factor - hcp, Tips for Calculating APF for hcp, Concepts and Terminology (6)

UNIVERSITY OF LEICESTER

Course Topics and Schedule

Lecture 12: The Physical Basis of Young's Modulus, Learning Objectives, Moduli of Crystals, Rubbers and the Glass Transition Temperature, Composite, Composite Terminology, Fibre-Reinforced Composites, Rule of Mixtures (1)
Lecture 13: Rule of Mixtures (2), Task, Concepts and Terminology (7)
Exercise Seminar 3
Lecture 14: The Yield Strength, Tensile Strength, Hardness and Ductility, Case Studies, Learning Objectives, How do Crystalline Materials Fail (1), Length Scales, Continuum Plastic Flow, Shearing, Shearing of a Ductile Polycrystal, Plastic Instability, What is Creep, Where is Creep Important
Lecture 15: How Does Creep Happen, Lattice Diffusion, Other Diffusion Paths, Creep Resistant Materials, Concepts and Terminology (8), Learning Objectives, How do Crystalline Materials Fail (2), Fracture Concepts, Fracture Jargon, Fracture Length Scales
Exercise Seminar 4

UNIVERSITY OF LEICESTER

Course Topics and Schedule

Lecture 16: Idealised Fracture Mechanisms, (More) Realistic Fracture Mechanisms, Brittle Fracture, Brittle Fracture - Amorphous Materials, Brittle Fracture - Zn at -120°C, Brittle Fracture - Al-Li 8090, Ductile Fracture by Tearing, Ductile Fracture - 'Pure' Cu, Fracture Mechanisms, Premature Failures, Brittle Fracture - Materials with Soft 'Voids', Brittle Fracture - Materials with Hard 'Inclusions', Ductility, Premature Failures, Concepts and Terminology (9)
Test
Lecture 17: Dislocations and Yielding in Crystals, Learning Objectives, The Structure of Metals and Ceramics (2), What Controls Mechanical Behaviour, Grains and Grain Boundaries, Grain Structure Examples, Imperfections, Hall-Petch Effect (1), Defects in Crystals
Lecture 18: Vacancies, Edge Dislocations, Edge Dislocation Movement, Slip Planes, Hall-Petch Effect (2), Strengthening Mechanisms - Grain Refinement, Limits of Hall-Petch Effect, Concepts and Terminology (10)

UNIVERSITY OF LEICESTER

Course Topics and Schedule

Exercise Seminar 5
Lecture 19: Revision of Theory
Lecture 20: Revision of Exercises

UNIVERSITY OF LEICESTER

Engineering Materials and Their Properties

UNIVERSITY OF LEICESTER

Module Overview

Exam Schedule

Final Exam: (Time and Location: To Be Announced)

Textbook

M.F. Ashby and D.R.H. Jones, Engineering Materials:

An Introduction to Properties, Applications and Design, Butterworth-Heinemann

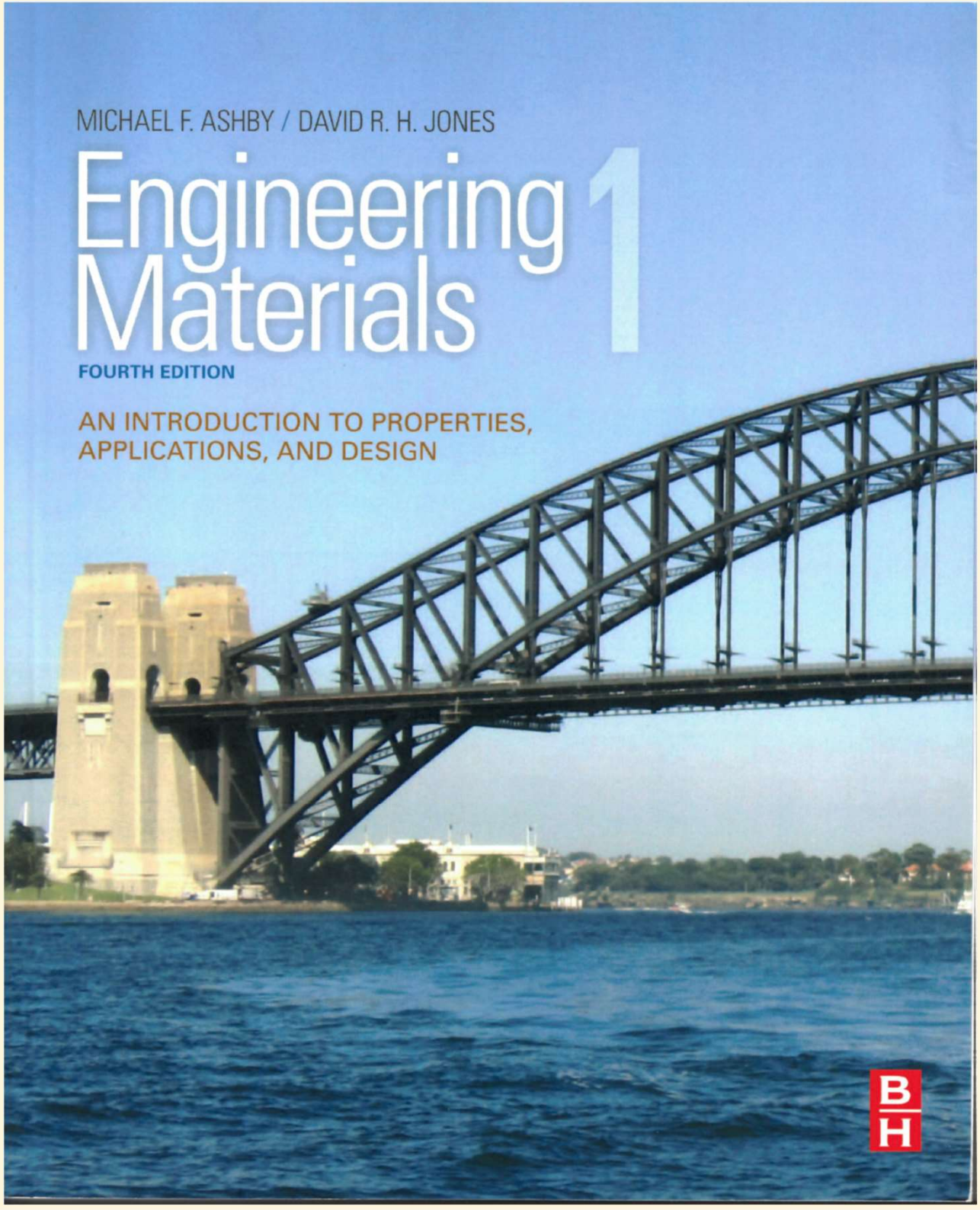
Reference Sources

Cambridge Engineering Selector (CES) EduPack software

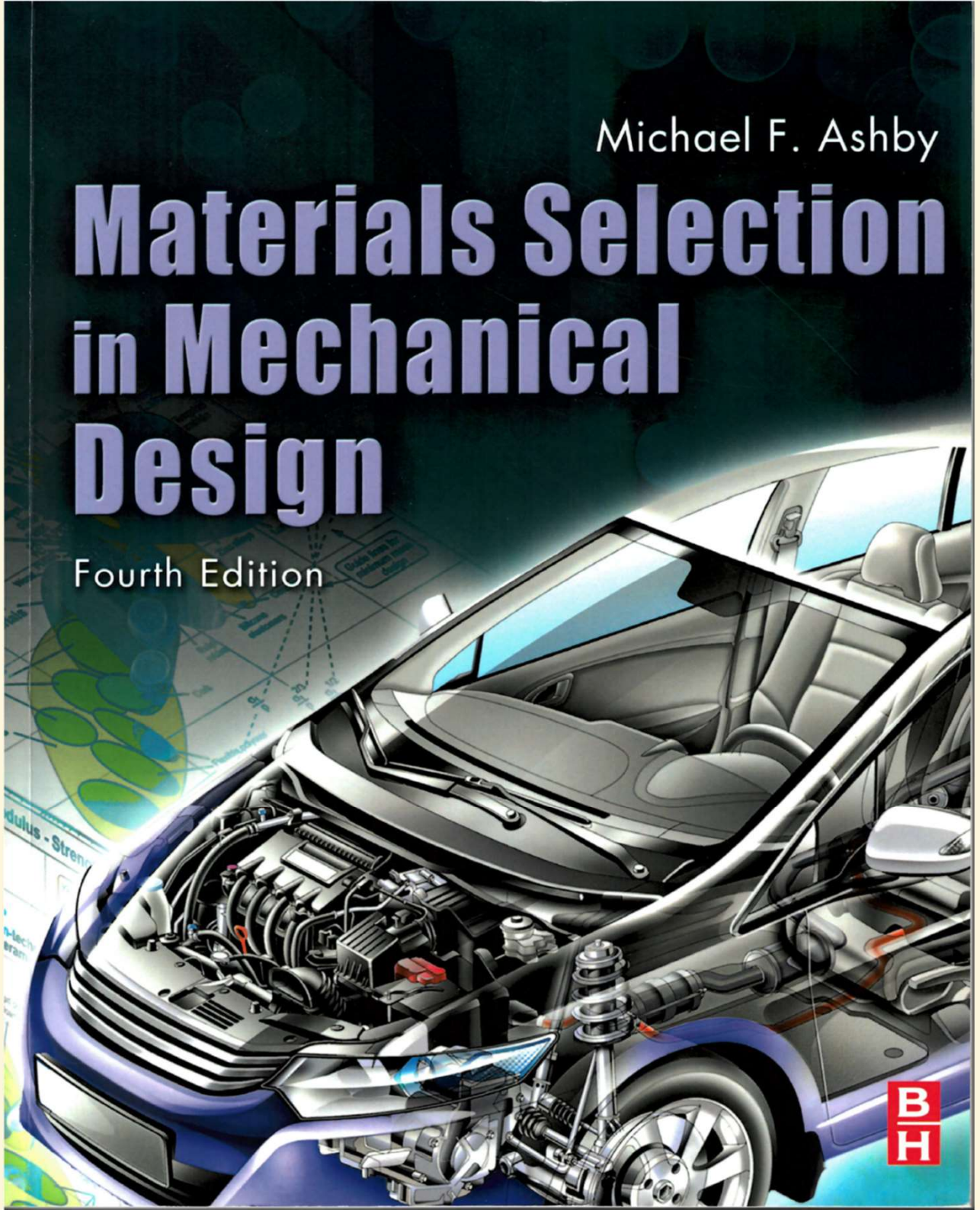
“Matter” on CFS & http://www.matter.org.uk/universities.htm

Doitpoms http://www.doitpoms.ac.uk/tlplib/index.php

UNIVERSITY OF LEICESTER



UNIVERSITY OF LEICESTER



UNIVERSITY OF LEICESTER

What this module is all about…

- The selection of Engineering **Materials**…
  - Metals, alloys; polymers; elastomers; glasses; ceramics
- …by their **properties**…
  - Stiffness, strength, toughness
- …their mechanical **behaviour** and **failure**…
  - Elastic, plastic, creep, brittle, ductile, fracture, fatigue and corrosion
- …and the **physical origins** of these properties and behaviours
  - Bonding, crystal structure, grain size, precipitates, alloying and dislocations!

UNIVERSITY OF LEICESTER

Module Aim

To provide an exposure to a wide range of materials and processing methods. This exposure should provide an insight into the effect of the processing of materials on the final structure and properties of engineering materials.

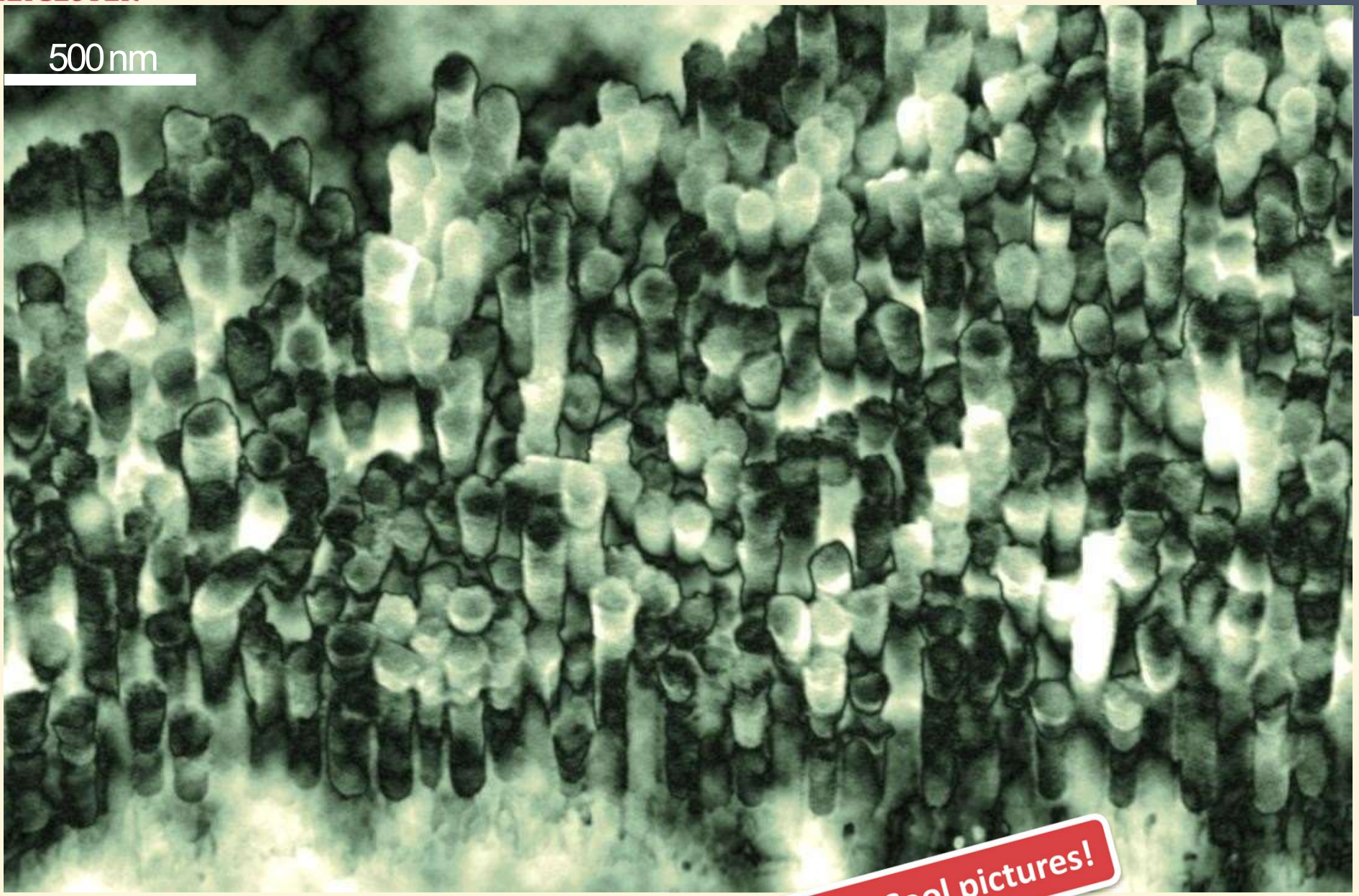
Why?

What use is this to me?

What help is it to an engineer?



UNIVERSITY OF LEICESTER



Why study materials?

Reason #1: Cool pictures!

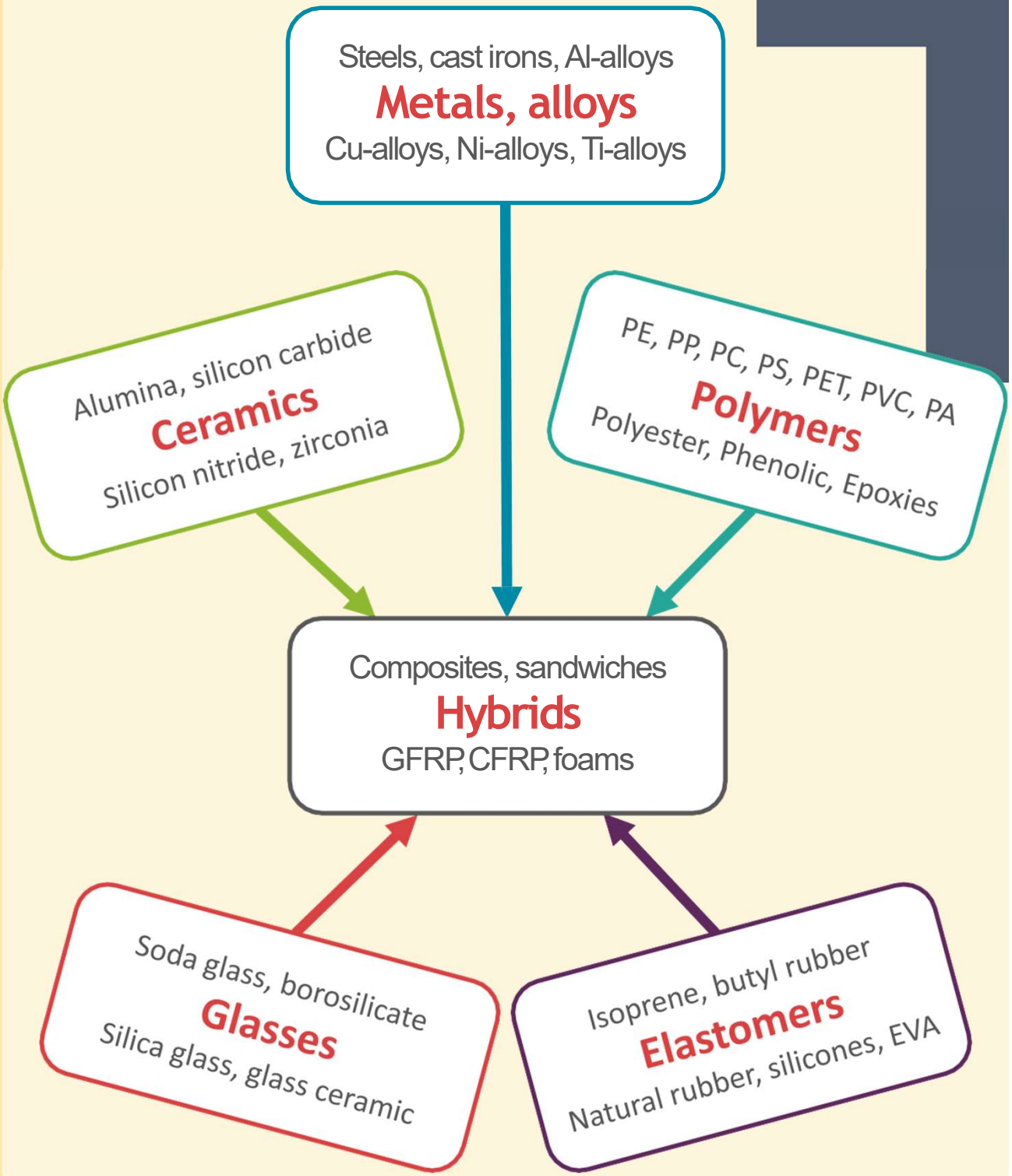
Nickel/Gold Nanowire Array

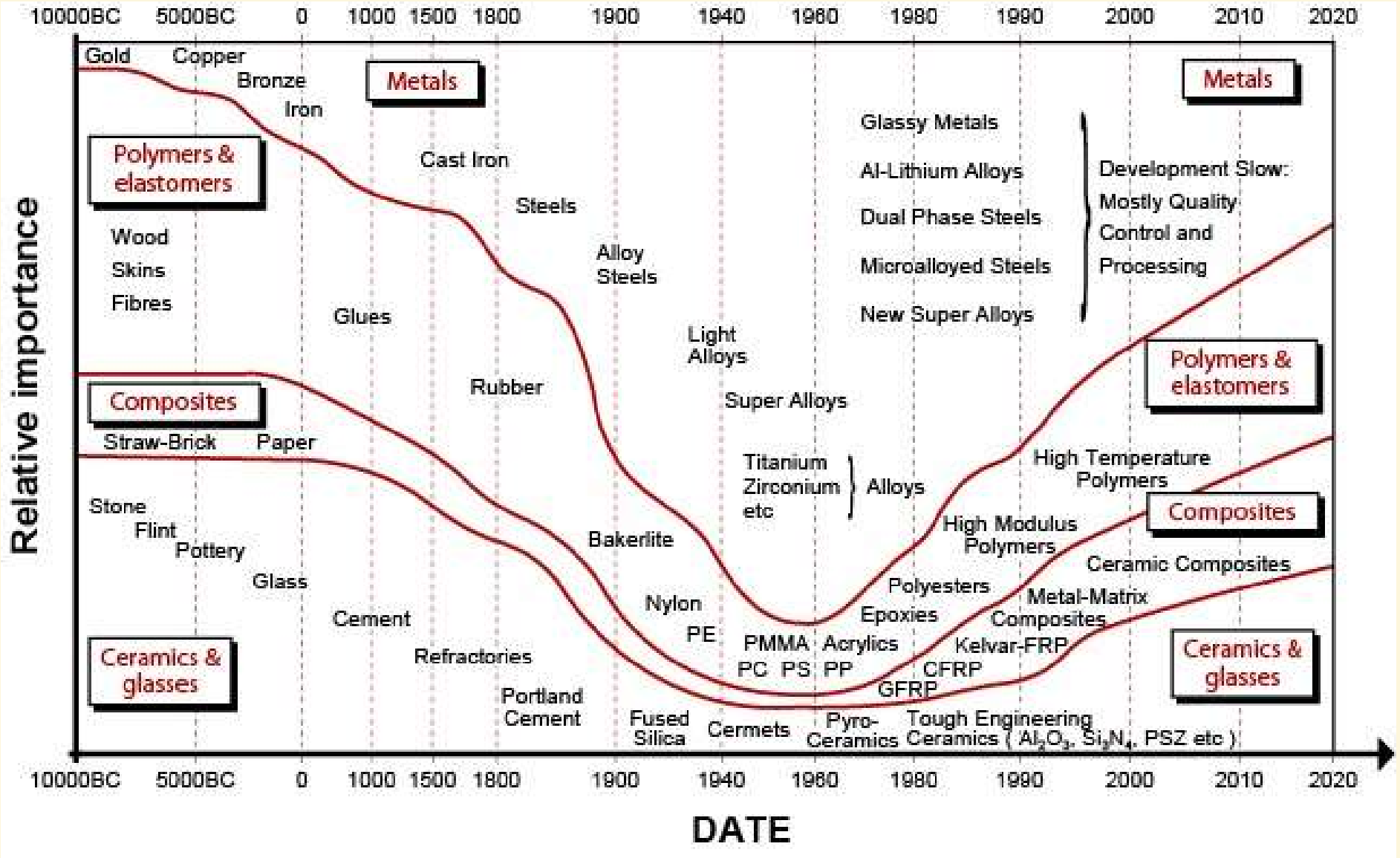
https://www.flickr.com/photos/40137058@N07/3783660261/in/photostream/

UNIVERSITY OF LEICESTER

The World of Materials

- >150,000 different materials available
- 5 families of materials
- Hybrids combine two or more of the others

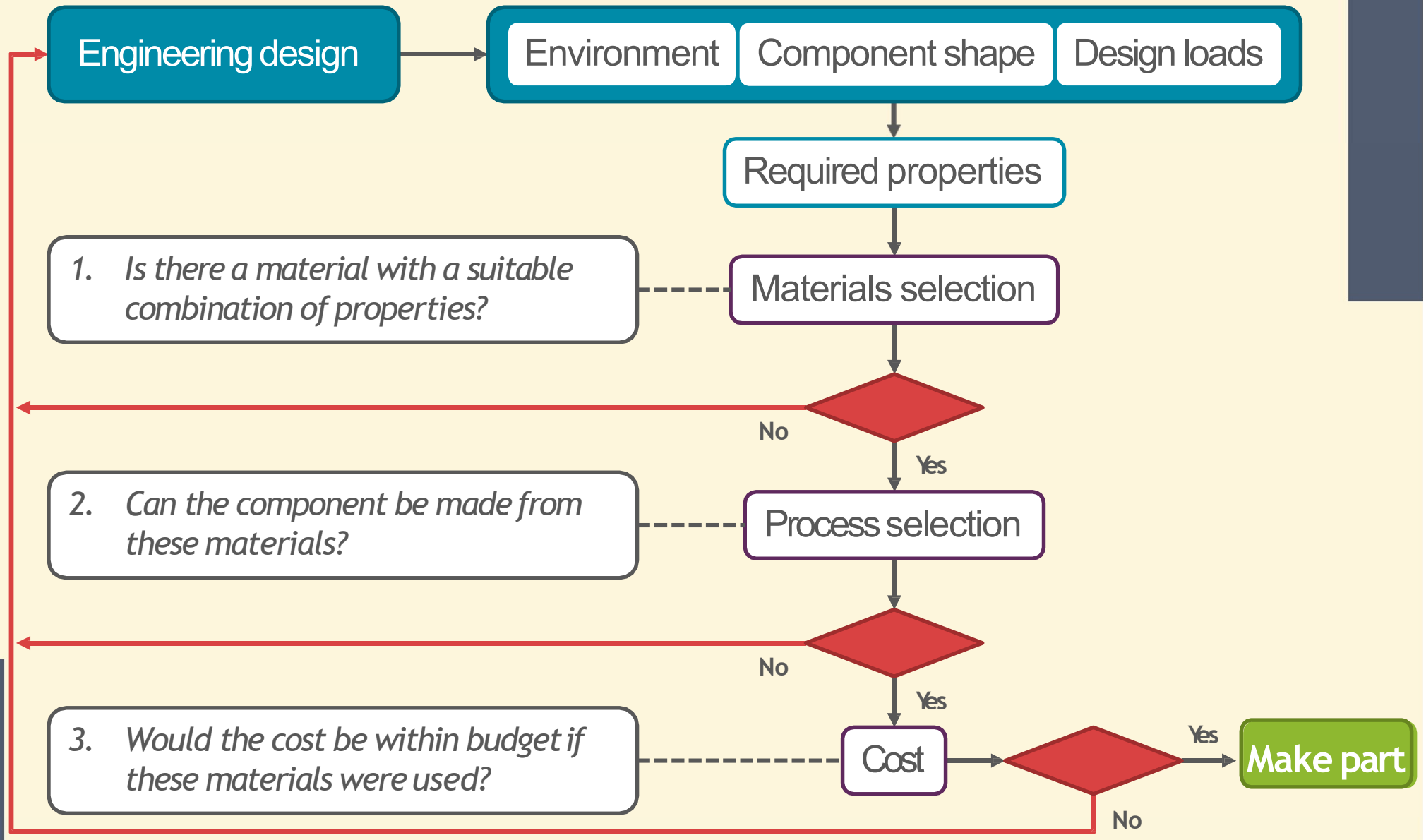




History – the evolution of materials

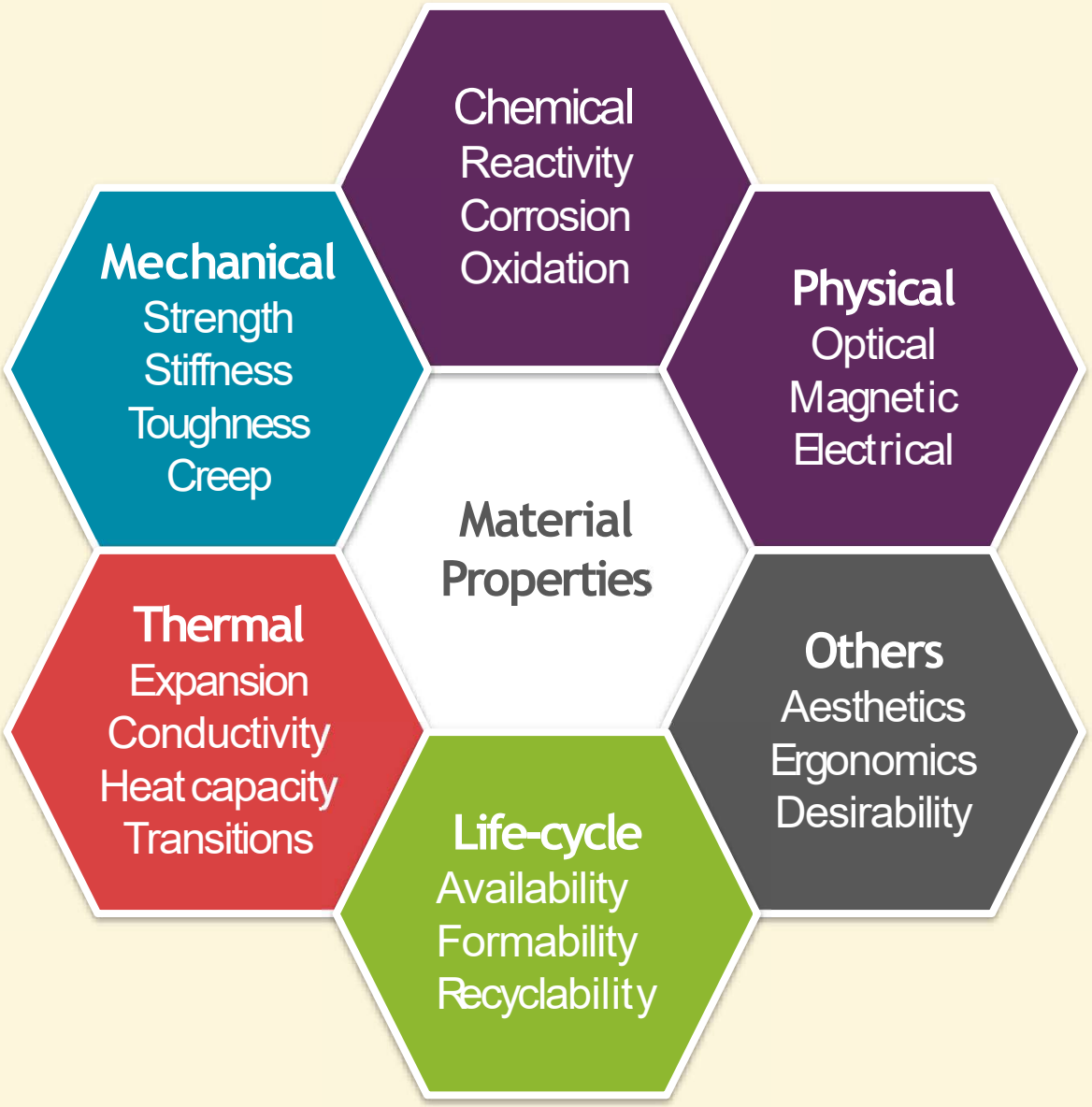
Image from: Ashby, M. F., (2011), *Materials Selection in Mechanical Design*

UNIVERSITY OF LEICESTER



Materials selection in design

UNIVERSITY OF LEICESTER



Properties of engineering materials



Pflaster/Pavement

https://www.flickr.com/photos/38365223@N03/7913024500/in/photostream/

Product 1: Paving (stone/concrete)

Non-critical — DESIGN REASONS — Critical


Photography by Design Services, University of Leicester

Product 2: Windows (glass)

Non-critical

DESIGN REASONS

Critical



Forth Rail Bridge, Scotland
http://commons.wikimedia.org/wiki/File:Forthbridge_feb_2013.jpg

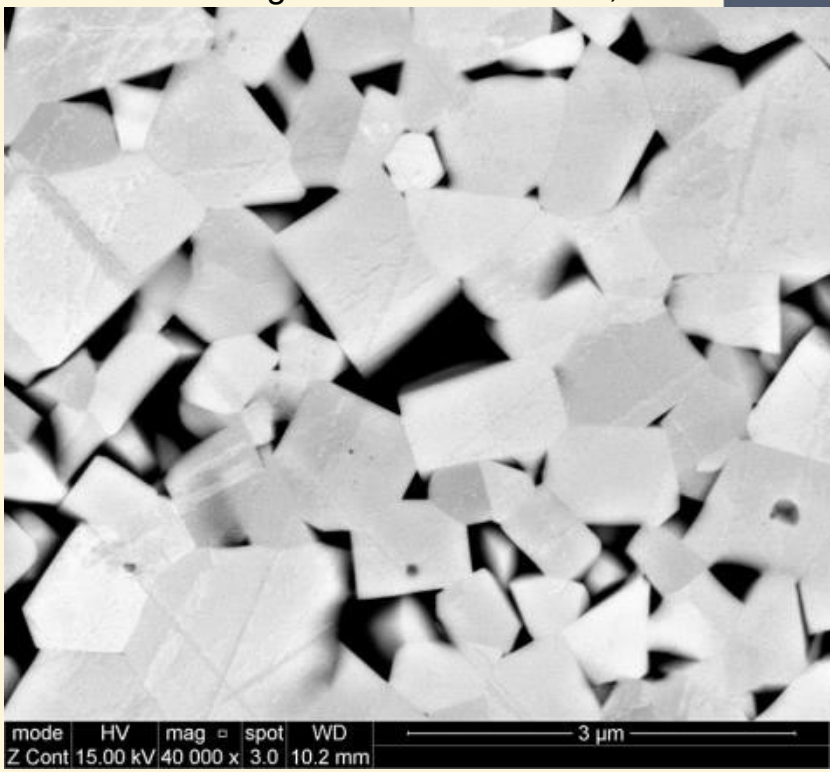
Product 3: Bridges (structural steel)

Non-critical — DESIGN REASONS — Critical



Hard Turning
http://www.machinery.co.uk/machinery-features/turning-lathes-one-hit-machining-miyano-hardinge/28045/

Cobalt-bonded tungsten carbide Thornton, 2014



Product 4: Cutting tools (tungsten carbides)

Non-critical — DESIGN REASONS — Critical

Fan blades

High-pressure turbine blades



Compressor blades

Low-pressure turbine blades

Rolls-Royce Trent engine
http://www.spaceflight.esa.int/impress/text/education/



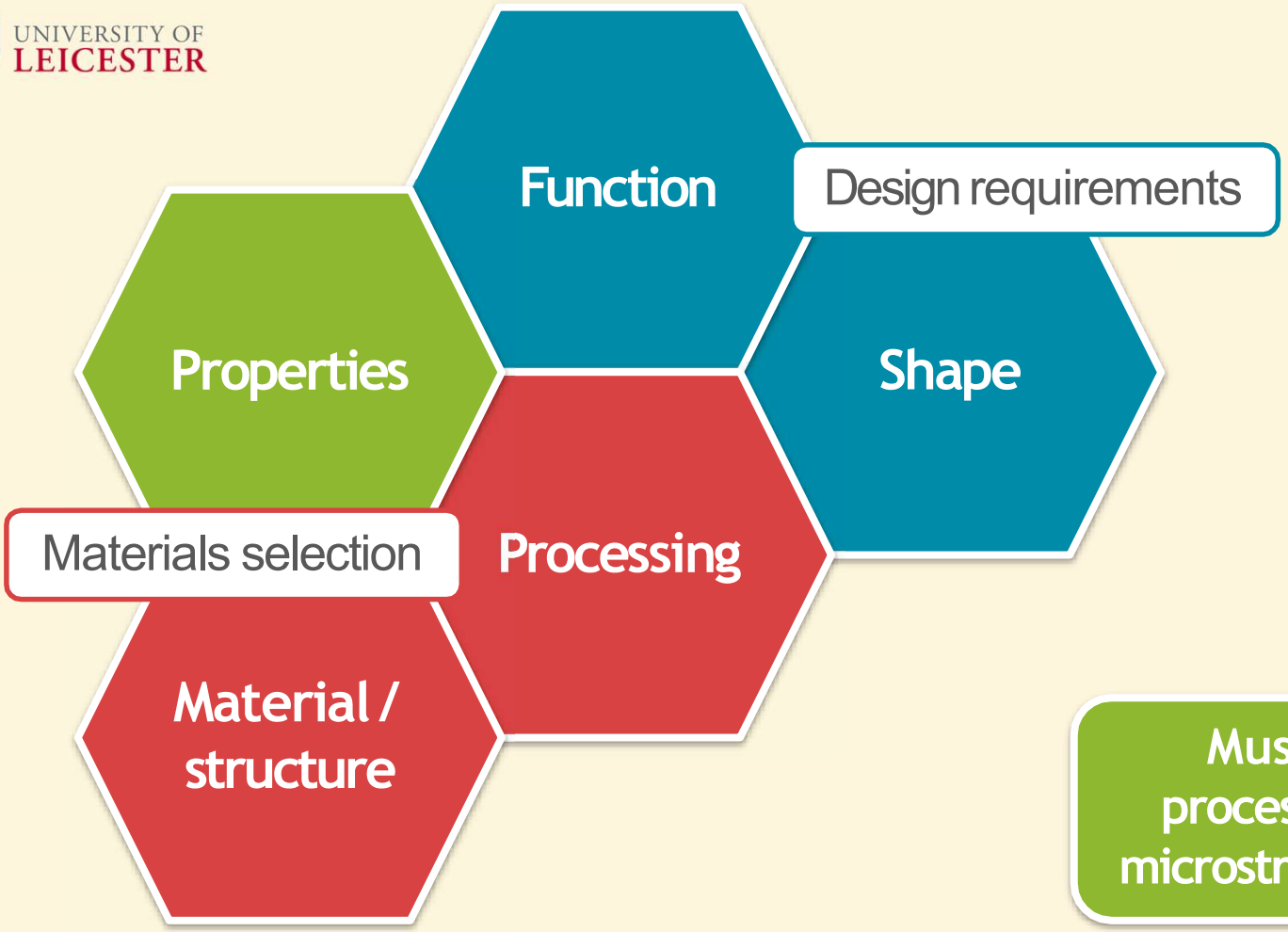
1600°C
10,000 rpm
Extracts 550 kW
from gasstream

High-pressure turbine blade
https://commons.wikimedia.org/wiki/File:GaTurbineBlade.svg

Product 5: Turbine blades (Ni-based superalloys)

Non-critical — DESIGN REASONS — Critical

UNIVERSITY OF LEICESTER



Must understand how processing history controls microstructures and properties

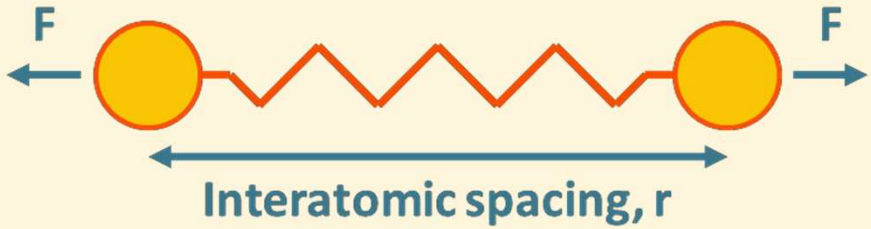
Three-way relationship

- Processing: shaping and tailoring material properties
- Structure: controlled by composition and processing
- Properties: controlled by microstructure

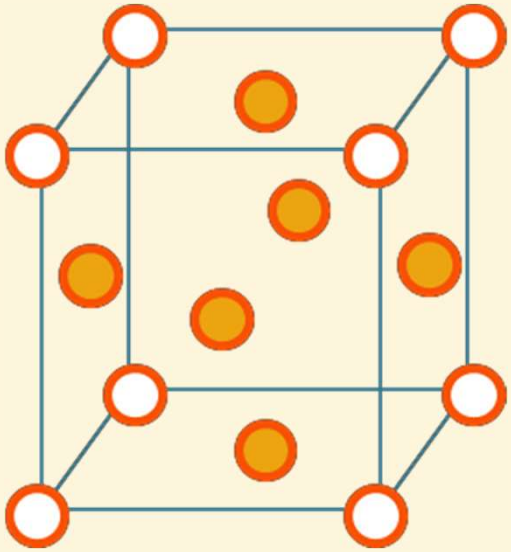
UNIVERSITY OF LEICESTER

Origins of material properties (1)

- Atomic **bonding**:
  - 'Strength of attraction' between atoms and chains of atoms



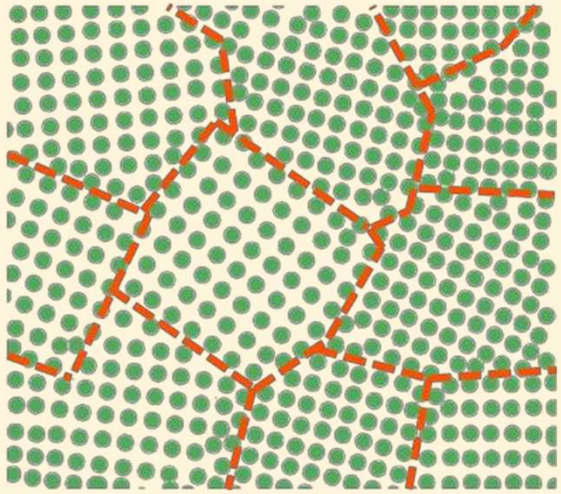
- Atomic **packing**:
  - Arrangement of atoms in solids; can be structured (e.g. metals) or un-structured (e.g. glasses)



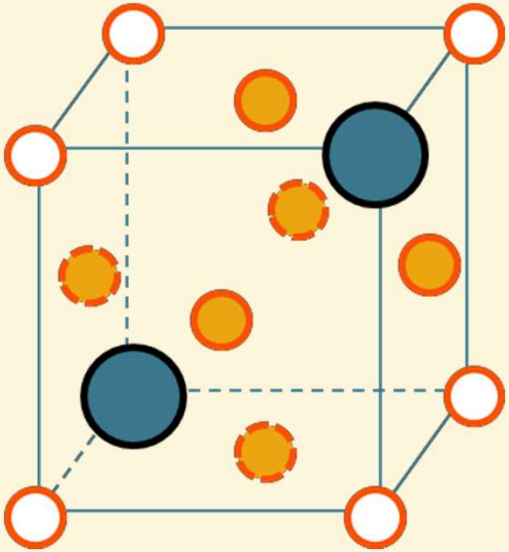
UNIVERSITY OF LEICESTER

Origins of material properties (2)

- **Grain size**:
  - Materials are rarely made up of a single homogeneous structure
  - Instead materials are formed of many irregular units or 'grains'



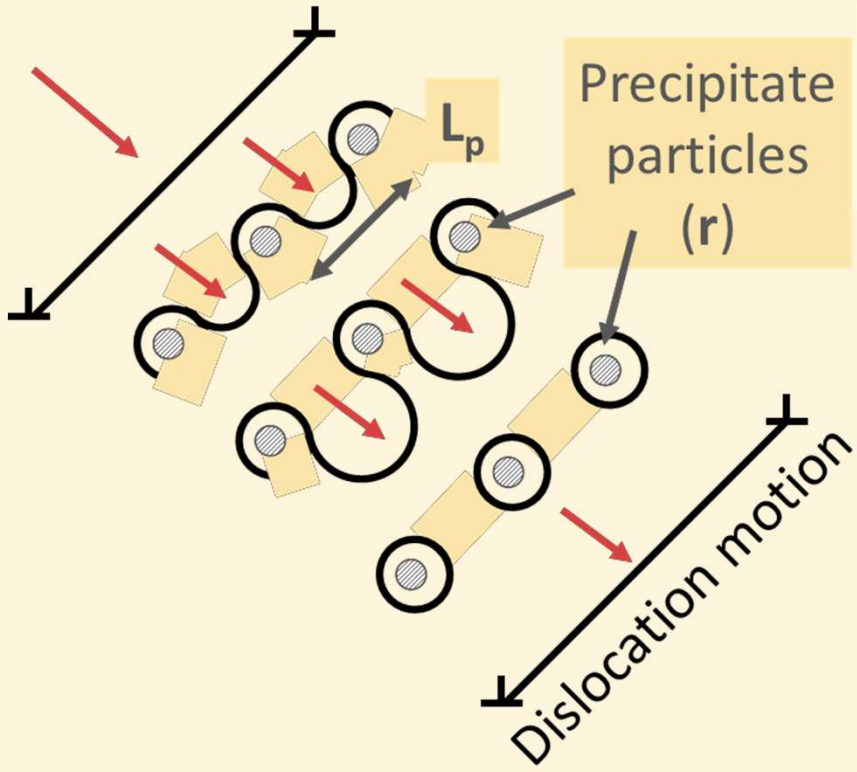
- **Alloying**:
  - Replacing atoms with those of another element
  - Results in 'lattice strain' that influences dislocation movement



UNIVERSITY OF LEICESTER

Origins of material properties (3)

- **Precipitates**:
  - Different (hard) chemical compounds that exist within a (soft) bulk material
  - Dislocations can become entangled around them, or have to cut through them



- **Dislocations**:
  - Atomic imperfections or defects in crystalline structures
  - Dislocation movement is the origin of plastic deformation

UNIVERSITY OF LEICESTER

How do we, as engineers, select materials?

- Materials selection process:
  1. **Translate** design requirements in constraints and objectives
  2. **Screen** materials to eliminate those that are not suitable
  3. **Rank** materials against performance metrics
  4. **Refine** selection using additional information

- Materials databases:
  - Online, e.g. MatWeb, Goodfellow, eFunda
  - Specialist software, e.g. Cambridge Engineering Selector EduPack

- **Next time, we will be learning how to apply the materials selection process and develop performance metrics**